

VIII. LINUX FILE SYSTEM ANALYSIS

Aim:

Perform real-time file system analysis on a Linux system to identify an attacker's artefacts.

Procedure:

1. Explore the Linux file system structure starting from the root (/) directory.
2. Identify and understand important directories such as /etc, /var, /home, and /bin.
3. Analyze file permissions and ownership using commands like `ls -l`.
4. Search for important or sensitive files using commands like `find` and `grep`.
5. Check log files and configuration files for useful information.
6. Document findings related to file structure, permissions, and potential security issues.

Tasks:

Task 1: Introduction

I'm ready to continue!

No answer needed

✓ Correct Answer

Task 2: Investigation Setup

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8}

✓ Correct Answer



Task 3: Files, Permissions and Timestamps

To practice your skills with the `find` command, locate all the files that the user `bob` created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3}

✓ Correct Answer



Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream

✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000

✓ Correct Answer

Task 4: Users and Groups

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d

✓ Correct Answer



What is the name of the group with the group ID of 46?

plugdev

✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree

✓ Correct Answer

Task 5: User Directories and Files

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b}

✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f}

✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000

✓ Correct Answer

Task 6: Binaries and Executables

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers

✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c

✓ Correct Answer

Task 7: Rootkits

Run `chkrootkit` on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh

✓ Correct Answer

Run `rkhunter` on the affected system. What is the result of the `(UID 0) accounts` check?

Warning

✓ Correct Answer

Task 8: Conclusion

Click and continue learning!

No answer needed

✓ Correct Answer

Result:

Successfully analyzed the Linux file system by identifying key directories, examining file permissions, and locating important files, helping to understand system structure and potential security concerns.